

GHASSEN ZRIGA

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EDUCATION

SUP'COM - Engineering Degree

Sept. 2023 – Dec. 2026

Telecommunications

Course material: Algorithms and Data Structures, Probability and Statistics, Telecommunication Systems, Digital Signal Processing, Networks, Communication Protocols.

IPEIN

Sept. 2021 – June 2023

Ranked 93rd out of 1785 in the national engineering entrance exam. Main courses: Linear Algebra, Analysis/Probability, Physics, Chemistry, Object-Oriented Programming (Python)

PROFESSIONAL EXPERIENCE

CRIL – Université d'Artois

May 2026 – present

Research internship - End of studies project (PFE)

- Experimenting with logical and neuro-symbolic frameworks to handle reasoning inconsistencies in LLMs, ensuring more robust and reliable decision-making processes.
- Using advanced logical reasoning into existing architectures to improve transparency, explainability, and fidelity of model decisions.

Sagemcom Tunisia

Oct. 2025 – February 2026

Tutored internship

- Developed an LLM-powered and GraphRAG-based solution to improve requirements quality in Jira/R4J through the construction of Knowledge Graphs on requirements bases (Neo4j) and semantic entity recognition for inconsistency detection, duplicate identification, and requirements description augmentation.
- Keywords: Requirements Engineering, NLP, LLM, Semantic Embeddings, Text Classification, R4J, Langchain, NetworkX

Huawei Technologies Tunisia

July – August 2025

Engineering internship

- Developed a reinforcement learning agent for multi-station LTE deployment with a custom Gym environment, implemented PPO as the learning algorithm with GNN actor and critic models, achieving 70% of exhaustive search capacity.
- Keywords: Reinforcement Learning, PyTorch, PyTorch Geometric, Gym, Rasterio, GNN, PPO

PROJECTS

- **LLM Financial Advisor:** Implemented a RAG-powered financial advisor chatbot based on a fine-tuned LLM (Llama 3 8b) with Unsloth, with a real-time financial news feature streaming pipeline using Alpaca's WebSockets stream and historical API to populate feature store in the RAG pipeline, fully served through an Azure container instance.
Keywords: Unsloth, Bytewax, Qdrant, Langchain, Weights & Biases, Azure, Alpaca API
- **Agentic Email Assistant:** Implemented a multi-agent system with CrewAI offering comprehensive email processing (sentiment analysis, task extraction, meeting suggestions), Gmail/Calendar integration, and smart attachment management with multilingual support for natural language queries. Engineered and tested prompts for each agent's specific task (Validator - Orchestrator - Task Extractor)
Keywords: CrewAI, FastAPI, Gemini API, Gmail API, React, Typescript, MySQL
- **IoT Irrigation Prediction System:** Designed and implemented an end-to-end IoT solution for intelligent irrigation and crop monitoring: sensor simulation, synthetic data generation and ingestion, preprocessing and feature engineering pipeline, integration and orchestration of an AI model (XGBoost for decision-making and water pump actuation), and dockerized modules ready for cloud deployment.
Keywords : Eclipse MQTT, NodeRED, Flask, FastAPI, Docker Compose, Scikit-learn, PostgreSQL

EXTRACURRICULAR EXPERIENCE

- Hackathons: Participated in several hackathons:
 - AI ANATOMY (GDC Sup'Com) – *4th place Kaggle Challenge*: Finetuned the Llama 3.1 8B model capable of generating precise answers from clinical case descriptions.
 - IndabaX Tunisia 2025 (IEEE Sup'Com SB) – *20th place/133*: XGBoost-based model for anomaly detection in telecom cell performance indicators.

LANGUAGES

French (bilingual)

English (C2)

Arabic (native)